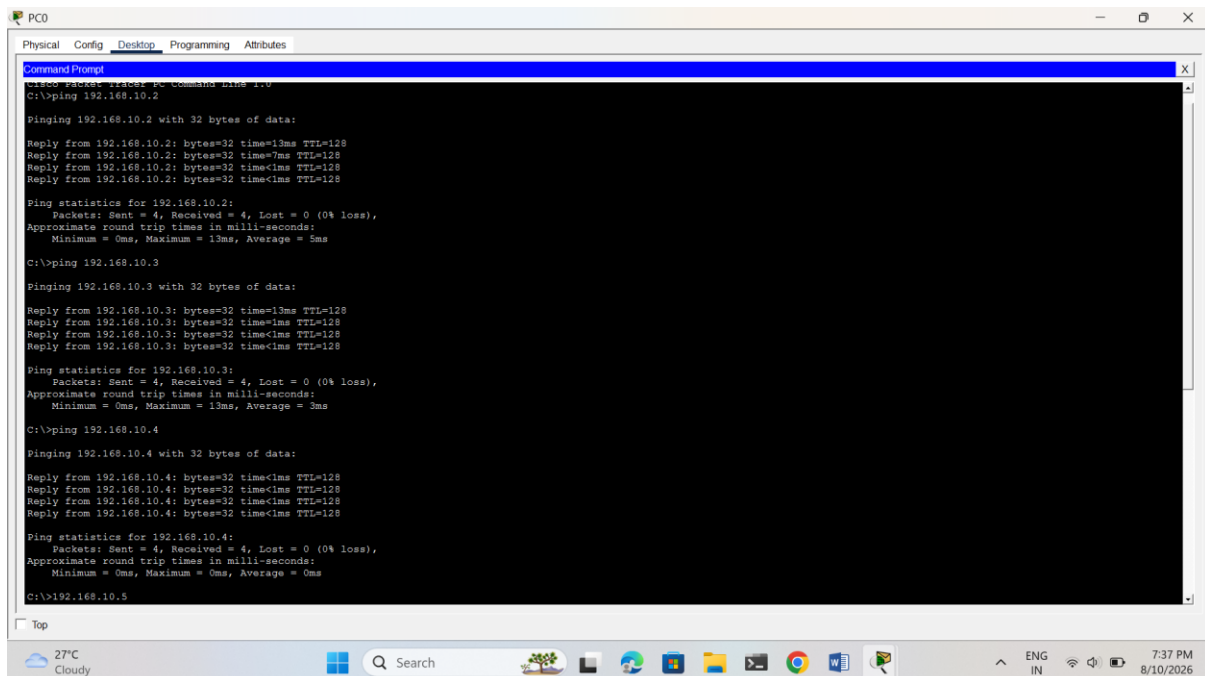


1. Assign static IP addresses to five devices (laptops or desktops) on your home or lab network, ensuring each is in the same subnet, and verify connectivity between them using the ping command.



The screenshot shows a Windows desktop environment with a window titled 'PC0'. The window has tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes', with 'Desktop' currently selected. Inside the window is a 'Command Prompt' application. The command prompt shows the following sequence of commands and outputs:

```
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time=13ms TTL=128
Reply from 192.168.10.2: bytes=32 time=7ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 13ms, Average = 5ms

C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time=13ms TTL=128
Reply from 192.168.10.3: bytes=32 time=1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 13ms, Average = 3ms

C:\>ping 192.168.10.4

Pinging 192.168.10.4 with 32 bytes of data:

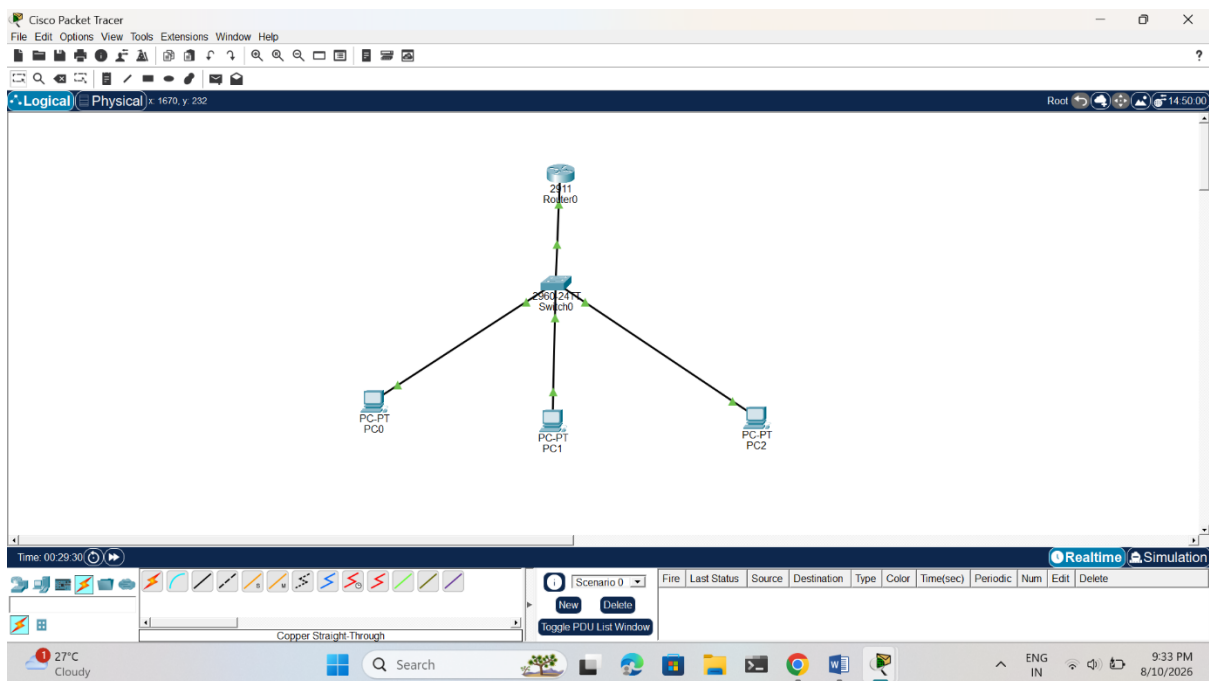
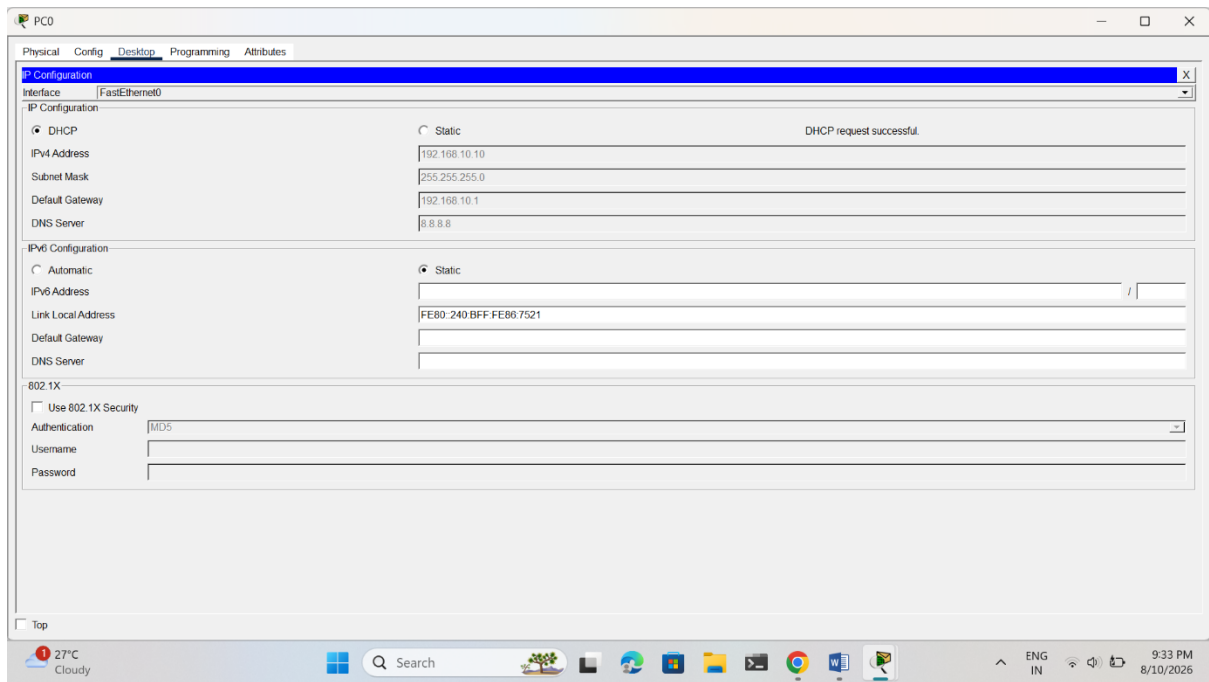
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

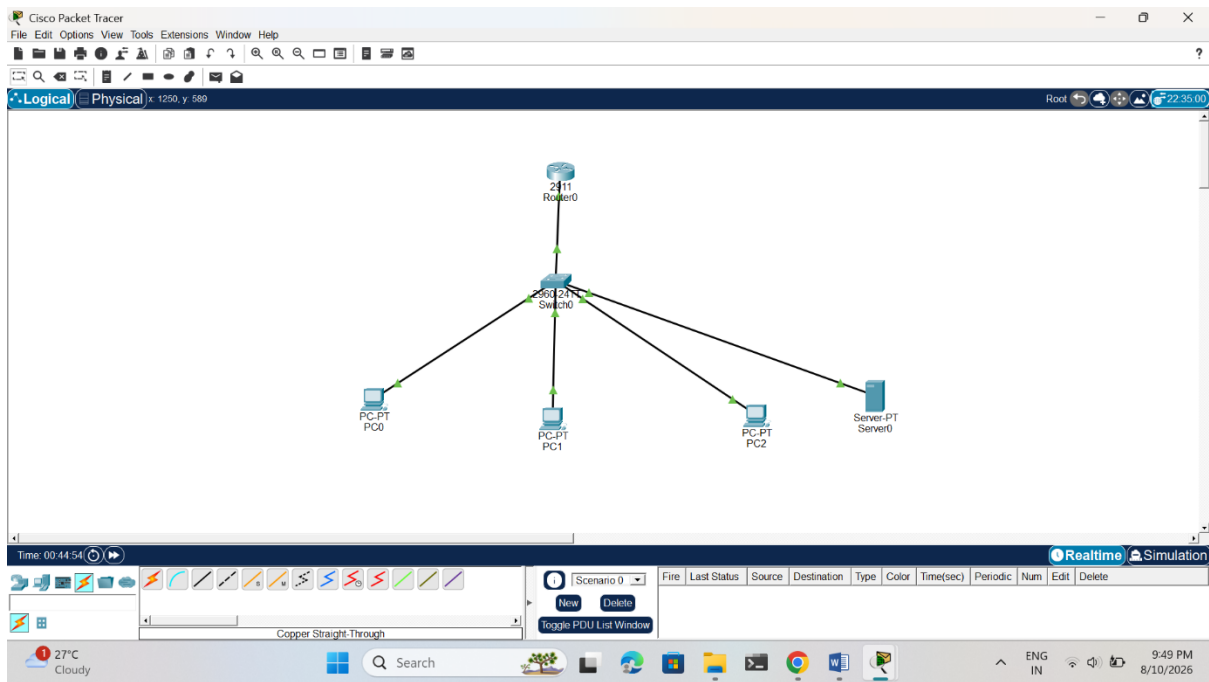
C:\>ping 192.168.10.5
```

The Windows taskbar at the bottom shows the system clock as 7:37 PM on 8/10/2026, and the weather as 27°C Cloudy.

2. Set up a DHCP server (using your router or software like Packet Tracer/GNS3/VirtualBox) to automatically assign IP addresses to at least three devices, and confirm that each device receives a unique IP from the correct range.



3. Configure a DNS service (using your router's DNS settings or a tool like dnsmasq) so that typing a custom hostname (e.g., musicplayer.local) in a browser on any connected device opens a specific local web page or device interface.
Hint: You can map hostnames to local IPs using the DNS service or by editing the hosts file for a quick test.



```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>nslookup musicplayer.local

Server: [192.168.10.20]
Address: 192.168.10.20
*** UnKnown can't find musicplayer.local: Non-existent domain.
C:\>
```

4. Create two VLANs (e.g., VLAN 10: 'Workstations', VLAN 20: 'Smart Devices') on your switch or simulation tool, assign at least two devices to each VLAN, and demonstrate that devices in the same VLAN can communicate, but not with devices in the other VLAN. Hint: Use ping or file sharing to test connectivity between devices in different VLANs.

```
C:\>ipconfig
```

```
FastEthernet0 Connection:(default port)
```

```
Connection-specific DNS Suffix...:
Link-local IPv6 Address.....: FE80::20B:BEFF:FEC5:2123
IPv6 Address.....: ::
IPv4 Address.....: 192.168.10.2
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
                        0.0.0.0
```

```
Bluetooth Connection:
```

```
Connection-specific DNS Suffix...:
Link-local IPv6 Address.....: ::
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
                        0.0.0.0
```